

1 The tree-sensitive lower-bound question

The transfer identity isolates an exact quartic defect of size

$$\frac{1}{64}$$

per level. After n levels this produces the deterministic discrepancy

$$\frac{n}{64}$$

in the fourth binomial moment.

However, every lower bound obtained after first averaging over histories appears to lose the essential causal information. In particular, the aggregate necessary conditions

$$Q^{*k} * B \succeq_{\text{st}} P^{*k}, \quad 1 \leq k \leq n,$$

do not force the root bill above its depth-one value.

Thus the obstruction is not merely distributional.

It is encoded in the branching structure of the exact tree.

To formulate the remaining problem, let

$$U_h(z) := \frac{1 - G_{B_h}(z)}{1 - z} = \sum_{j \geq 0} \Pr(B_h > j) z^j$$

be the tail generating function of the reserve law at history h . Then

$$\|U_h\|_{\ell^1} = \mathbb{E}B_h.$$

Using

$$G_P(z) - G_Q(z) = \frac{(1 - z)^4}{64},$$

the exact zero-repair recursion is equivalent to

$$\boxed{G_Q(z)U_h(z) - \sum_{x \in \mathcal{A}} P(x)z^x U_{hx}(z) + \frac{(1 - z)^3}{64} = 0.} \quad (1)$$

Thus the quartic discrepancy of the original laws becomes a *cubic forcing term* in tail coordinates.

The bill itself is the total tail mass.

This suggests the following question.

[Tree-sensitive dual potential] Does there exist a finite collection of nonnegative functionals

$$\Lambda_s$$

on probability laws on $\mathbb{Z}_{\geq 0}$, indexed by a finite auxiliary state s , together with constants

$$c > 0, \quad C < \infty,$$

such that the following two properties hold?

(i) Mean control. For every law B of finite mean,

$$\boxed{0 \leq \Lambda_s(B) \leq C \mathbb{E}B \quad \text{for every } s.} \quad (2)$$

(ii) **Positive tree drift.** For every exact zero-repair node

$$Q * B = \sum_{x \in \mathcal{A}} P(x) \delta_x * B_x$$

and every auxiliary state s ,

$$\Lambda_s(B) \geq \sum_{x \in \mathcal{A}} P(x) \Lambda_{\tau(s,x)}(B_x) + \frac{c}{(1 + \mathbb{E}B)^3}, \quad (3)$$

where

$$\tau(s, x)$$

is a fixed finite-state update rule independent of the horizon?

The exponent 3 is not arbitrary.

Equation (1) contains the universal forcing

$$\frac{(1-z)^3}{64}.$$

If one tests this forcing against a bounded weight varying on a spatial scale L , its contribution is governed by a third discrete difference and therefore naturally has size

$$L^{-3}.$$

Since the relevant scale of the reserve is expected to be comparable to its mean, the first scale-invariant candidate is precisely

$$\rho(m) \asymp m^{-3}.$$

The importance of Problem 1 is that even this apparently weak drift would already decide the logarithmic question.

Proposition 1 (A cubic tree drift excludes logarithmic growth). *Assume that functionals satisfying (2) and (3) exist. Then there is a constant $c_0 > 0$ such that*

$$\mathcal{B}_n \geq c_0 n^{1/4}$$

for all sufficiently large n . In particular,

$$\mathcal{B}_n \neq O(\log n).$$

Proof. Let

$$H_t = (X_1, \dots, X_t), \quad M_t := \mathbb{E}B_{H_t},$$

along a random branch with

$$X_1, X_2, \dots \stackrel{\text{iid}}{\sim} P.$$

Taking means in the exact zero-repair recursion and using

$$\mathbb{E}P = \mathbb{E}Q$$

gives

$$\mathbb{E}[M_{t+1} \mid H_t] = M_t.$$

Hence (M_t) is a nonnegative martingale and

$$\mathbb{E}M_t = M_0.$$

Iterating (3) down an exact depth- n tree gives

$$\Lambda_{s_0}(B_\emptyset) \geq \mathbb{E}\Lambda_{s_n}(B_{H_n}) + c \sum_{t=0}^{n-1} \mathbb{E} \frac{1}{(1 + M_t)^3}.$$

Since all potentials are nonnegative,

$$CM_0 \geq c \sum_{t=0}^{n-1} \mathbb{E} \frac{1}{(1 + M_t)^3}.$$

The function

$$x \mapsto \frac{1}{(1 + x)^3}$$

is convex on $[0, \infty)$, so Jensen's inequality yields

$$\mathbb{E} \frac{1}{(1 + M_t)^3} \geq \frac{1}{(1 + \mathbb{E}M_t)^3} = \frac{1}{(1 + M_0)^3}.$$

Therefore

$$CM_0 \geq \frac{cn}{(1 + M_0)^3},$$

or equivalently

$$M_0(1 + M_0)^3 \geq \frac{c}{C} n.$$

Thus

$$M_0 \geq c_0 n^{1/4}$$

for some $c_0 > 0$. Taking the infimum over all exact depth- n trees proves the claim. $\square$

Remark 2 (The stronger square-root target). *The numerical finite-memory calculations suggest a substantially stronger law,*

$$\mathcal{B}_n \asymp \sqrt{n}.$$

The corresponding local drift would have order

$$\frac{1}{1 + \mathbb{E}B}$$

rather than

$$\frac{1}{(1 + \mathbb{E}B)^3}.$$

Indeed, if one could replace (3) by

$$\Lambda_s(B) \geq \sum_x P(x) \Lambda_{\tau(s,x)}(B_x) + \frac{c}{1 + \mathbb{E}B},$$

the same martingale–Jensen argument would give

$$\boxed{\mathcal{B}_n \gtrsim \sqrt{n}.}$$

The cubic version should therefore be regarded as the first rigorous target, not necessarily the sharp one.

Remark 3 (Why the functional must be tree-sensitive). *A functional depending only on the aggregate law*

$$Q^{*k} * B_\emptyset$$

cannot exploit the information discarded when the exact tree is averaged over histories.

The desired potential must therefore be applied before this averaging takes place.

Equivalently, it must detect how the universal cubic forcing

$$\frac{(1-z)^3}{64}$$

is distributed among the different children

$$B_{hx}$$

of an exact node.

This is the missing information in the aggregate moment and stochastic dominance relaxations.

[Sharp form] Can one find a finite-state family of bounded tail potentials for which the exact tree recursion forces

$$\Lambda_s(B) - \sum_x P(x) \Lambda_{\tau(s,x)}(B_x) \gtrsim \frac{1}{1 + \mathbb{E}B} ?$$

A positive answer would imply

$$\mathcal{B}_n \gtrsim \sqrt{n}.$$

More modestly, can one prove any estimate

$$\Lambda_s(B) - \sum_x P(x) \Lambda_{\tau(s,x)}(B_x) \gtrsim \frac{1}{(1 + \mathbb{E}B)^p}$$

with finite p ?

Such an estimate would imply

$$\mathcal{B}_n \gtrsim n^{1/(p+1)},$$

and hence would already rule out logarithmic growth.

Thus the essential question is not yet the exact value of the exponent.

It is whether the quartic forcing can be converted into *any positive tree-sensitive drift whose decay in the bill is only polynomial.*

Question to the reader.

Can one construct a bounded, tree-sensitive dual potential which sees the cubic tail forcing

$$\frac{(1-z)^3}{64}$$

before the branching information is averaged away?

Equivalently: can the exact quartic defect be converted into a uniform positive drift along every admissible zero-repair tree, with cost controlled only by the current expected reserve?

Any polynomial drift

$$\rho(m) \gtrsim (1+m)^{-p}, \quad p < \infty,$$

would already imply a polynomial lower bound

$$\mathcal{B}_n \gtrsim n^{1/(p+1)}$$

and would therefore settle the logarithmic-versus-polynomial dichotomy.